

2. Iris Dataset Analysis Using a Voronoi Classifier

October 7, 2024

The Iris data set is a widely distributed collection of data pertaining to three common varieties of Iris: *I. setosa*, *I. versicolor*, and *I. virginica*. For each subspecies, four features were measured: sepal length, sepal width, petal length, and petal width.

Results: First, a linear classification of the setosa vs versicolor was implemented. The sepal length is plotted against sepal width, petal length, and petal width. Then the sepal length vs sepal width is used to classify using both the Gaussian and Sinc kernels. The following is a table of errors determined for each plot.

Table 1: Classification Error on Test Data for Each Plot (*I. setosa* vs *I. versicolor*)

Plot No.	Percent Error
1	0
2	5
3	0
4	0
5	10